

Hanxiao Wu

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Research Interests

Polar geophysics; integrated seismic and radar imaging of the subglacial environment; Antarctic subglacial structure, hydrology, and ice-bed interactions; geophysical inversion and crustal structure.

Education

Stony Brook University 2021–2027 (expected)

Ph.D. in Geophysics, Advisor: Weisen Shen

Dissertation: From Seismic Observations to Subsurface Structure and Physical Properties: Applications to Continental and Subglacial Environments.

University of Science and Technology of China 2018–2021

M.S. in Geophysics, Advisor: Huajian Yao

Thesis: Research on Joint Seismological Inversion Methods for the 1-D Layered Crustal Structure

University of Science and Technology of China 2014–2018

B.S. in Geophysics

Research Experience

Graduate Research Assistant 2021–Present

Stony Brook University, Advisor: Weisen Shen

Antarctic Subglacial Geophysics

- Use dense seismic observations to investigate the structure, physical properties, and hydrological state of subglacial sediments beneath the Antarctic Ice Sheet.
- Use independent radar and magnetotelluric observations to provide complementary evidence for the interpretation of seismic constraints on subglacial conditions.

Continental Crustal Structure and Seismic Inversion

- Develop joint-inversion methods that combine complementary seismic observations to constrain crustal structure and reduce parameter trade-offs.
- Apply these methods to large-scale USArray datasets to investigate crustal structure and its spatial variations across the continental United States.

Research Assistant 2018–2021

University of Science and Technology of China

- Developed joint-inversion methods for layered azimuthal anisotropy to constrain depth-dependent seismic anisotropy and crustal structure.

Polar & Geophysical Field Experience

Antarctica Polar Array — South Pole, Antarctica 2024–2026

Two austral field seasons; Field Team Lead (2025–2026).

Large-N nodal array deployment and recovery, broadband station servicing, array/profile design, and remote-field operations.

Seismic Field Deployment — Kenya 2024

D deployed ~ 80 SmartSolo nodes; contributed to survey design and field data acquisition.

Publications

Peer-Reviewed Publications

1. **Wu, H.**, Sui, S., and Shen, W. (2024). Incorporating H- κ Stacking with Monte Carlo Joint Inversion of Multiple Seismic Observables: A Case Study for the Northwestern US. *Journal of Geophysical Research: Solid Earth*, 129(7).

2. Yang, Y., Yao, H., **Wu, H.**, Zhang, P., and Wang, M. (2020). A New Crustal Shear-Velocity Model in Southwest China from Joint Seismological Inversion and Its Implications for Regional Crustal Dynamics. *Geophysical Journal International*, 220(2), 1379–1393.

Manuscripts Under Review

1. **Wu, H.**, et al. *A Water-Rich Sedimentary Aquifer System Beneath the South Pole Controls Basal Conditions of the Antarctic Ice Sheet*. Manuscript under review at *Science*.
2. **Wu, H.**, et al. *A Comprehensive Seismic Array in the South Pole, East Antarctica* Manuscript under review at *Seismological Research Letters*.

Manuscripts in Preparation

1. **Wu, H.**, et al. *A Continental-Scale Crustal Model of the United States from Joint Inversion of Multiple Seismic Observables* In preparation.
2. **Wu, H.**, et al. *Diffusion Models for Automated Quality Control of Receiver Functions* In preparation.

Talks & Presentations

1. **Wu, H.** and Shen, W. (2026). “Large-N Seismic Nodes and Broadband Observations at the South Pole: Enabling High-Resolution Cryoseismology in Extreme Polar Environments.” *Seismological Society of America Annual Meeting*, Pasadena, CA. Oral Presentation.
2. **Wu, H.** (2025). “Revealing the Hidden Water Beneath the South Pole.” Invited Talk, *Geophysics Lunch Bunch*, Brown University.
3. **Wu, H.**, Shen, W., and Kumar, A. (2025). “Revealing the Hidden Water beneath the South Pole.” *AGU Fall Meeting*, NS31A-05.
4. **Wu, H.** (2025). “Partial Melting in the Western US? Results of a Novel Joint Inversion Method for Layered Crustal V_P/V_S .” *Interior of the Earth Gordon Research Seminar and Gordon Research Conference*, South Hadley, MA. Poster Presentation.
5. **Wu, H.**, Shen, W., Lin, F.-C., and Sui, S. (2023). “A Novel Joint Inversion Method for Layered Crustal V_P/V_S Ratio: Incorporating Rayleigh Wave Local Amplification.” *AGU Fall Meeting*, S23F-0423.
6. **Wu, H.** and Shen, W. (2022). “Inferring the Depth-Dependent V_P/V_S of the Crust: Embedding H- κ Stacking into Monte Carlo Joint Inversion of Multiple Data Sets.” *AGU Fall Meeting*, S42B-05.
7. **Wu, H.**, Shen, W., and Lin, F.-C. (2022). “Monte Carlo Joint Inversion of Crustal Architecture: Incorporating Local Amplifications.” *SAGE/GAGE Community Science Workshop*.
8. **Wu, H.**, Shen, W., and Schulte-Pelkum, V. (2022). “The Apparent Seasonal Variations in Seismic Observations Made in Antarctica and Other Polar Regions.” *SCAR INSTANT PSE Workshop*.

Industry Research Experience

Geophysical Research Intern

Summer 2026

TGS

Developed and productionized Stein variational gradient descent (SVGD) workflows for uncertainty quantification in 3-D/4-D acoustic full-waveform inversion, including scalable HPC/GPU implementation and uncertainty analysis.

Teaching & Mentoring

Research Mentoring

- Mentored two high school students in independent seismology research projects on Antarctic surface-wave imaging and upper-mantle anisotropy; both projects resulted in student-led presentations at the AGU Fall Meeting (2024 and 2025).

Teaching Assistant, Stony Brook University

- Introduction to Geophysics, Fall 2021 and Fall 2022
- Introduction to Hydrology, 2022

Teaching Assistant, University of Science and Technology of China

- Geophysical Inversion Theory, Fall 2019

Awards & Recognition

- PSECCO Travel Grant, Seismological Society of America Annual Meeting, 2026.
- Antarctica Service Medal, National Science Foundation, 2025.
- Graduate Excellence in Teaching Award, Stony Brook University, 2022.
- SAGE/GAGE Workshop Student Travel Scholarship, 2021.

Technical Skills

Seismology & Geophysics: Receiver functions, surface-wave dispersion, ambient noise, H/V analysis, seismic anisotropy, joint inversion, Machine Learning.

Computational: Python, C/C++, Fortran, Bash; Monte Carlo and Bayesian inversion; HPC and GPU computing; large-scale seismic data processing.

Field: Large-N nodal seismic arrays, broadband seismometers, seismic survey and array design, Antarctic and remote-field operations.